



MAID

What is MAID?

MAID stands for Massive Array of Idle Disks. This is a technology that promises to provide massive scaleable storage at a very low cost.

Where will it be used?

MAID will be used where there is a need to archive or back up data amounting to terabytes. It will also be used in places where large amount of archived data needs to be accessed quickly.

Who is selling MAID solutions?

Though the concept has been around some time now, only one company—Copan Systems—offers the solution.

What was the alternative to this technology?

The older alternative to MAID is archiving using a tape drive. There are many drawbacks to tape, the main one being slow access speeds. Also, when a large amount of data in image form has to be stored on a tape drive that has less storage capacity than the image, the image has to be split—which is cumbersome and also takes a lot of time. Sometimes it takes days to access the archived data stored in tapes, discouraging active usage.

How scalable is MAID?

MAID is an array of hard drives combined to work as a single point of storage. Scalability is therefore not an issue. The smallest of Copan's systems is 28 terabytes, and the largest is 200 terabytes. The space used for the largest-capacity array Copan produces is just one square metre, making for space-saving.

Why MAID?

MAID consumes less power and produces less heat, since when the data is not accessed on the hard disks, the disks go into idle mode—making it energy-efficient. MAID allows one to store data in both structured (SQL, MS Access) and unstructured data formats (Word, PowerPoint).

FLASH IN HARD DRIVES

Faster Bootups, Longer Battery Life

Microsoft and Samsung first demonstrated a hard drive with Flash memory about a year ago. This month, they will showcase the technology again in a form they claim will be ready to come to market. The system will be demonstrated at the WinHEC (Windows Hardware Engineering Conference), which started 24 May. However, the OS code that supports the Flash package will only ship with Windows Vista.

The unit that Samsung showed off last year had 128 MB of Flash, which was used to cache startup data and other frequently-accessed information, both to make the loading of the data into RAM faster and to boost laptop battery life.

When Vista arrives, Intel will be promoting its own, related technique, which incorporates the Flash on an add-in card rather than putting it in the hard drive. This approach allows the Flash to be replaced easily and will keep hard drive prices down, but will raise the system price. In fact, Intel, Hynix, Micron, Sony and other companies have founded the Open NAND Flash Interface (ONFI) working group, which will develop a specification that might make it easier to incorporate Flash memory in laptops.

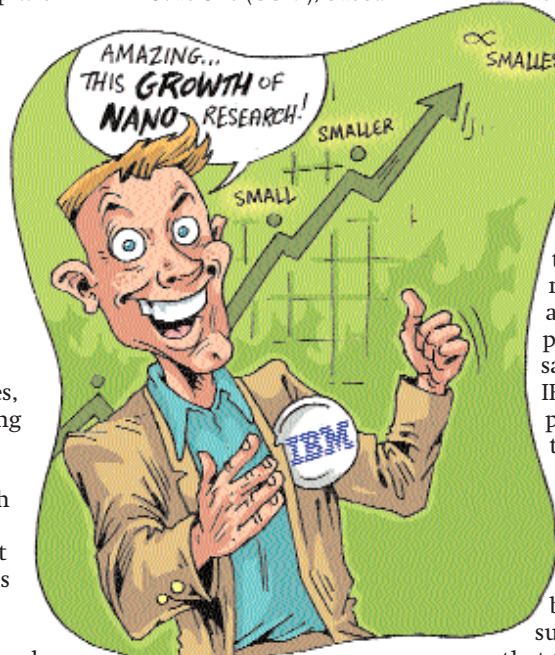
Intel had actually announced the formation of ONFI at its Developer Forum in March. It showed off an add-in card codenamed "Robson" designed to be

fitted into a notebook to accelerate the boot process and the loading of applications. ONFI will define a standardised interface by which the Flash chips mounted on the card communicate with the card's controller chip. The initial specification is expected to be completed in the second half of this year.

IBM'S BLUEGENE

Nanotech Gets A Boost

IBM is to back a research centre being developed at Rensselaer Polytechnic Institute (RPI). The \$100 million Computational Center for Nanotechnology Innovations (CCNI), based



at Rensselaer Technology Park in Troy, New York, will further research into shrinking electronic devices including computer chips. An IBM BlueGene supercomputer, yet to be built, will power the effort. It will be able to perform 70 trillion calculations per second (70 teraflops).

Herb Schultz, marketing manager for BlueGene at IBM, explained, "To do simulations on how molecules react, which is the essence of

nanotechnology, researchers need the computing power that BlueGene delivers." Underlining the growing importance of nanotech, Schultz went on, "There are so many applications for this that will be emerging in the next decade." He also hinted that using BlueGene to work on computational problems such as ways to extend Moore's Law could open new opportunities for a range of industries.

Among the technologies that might benefit from the centre are super-fast nanocapacitors and nanoscale memory cells. These would have applications in Flash-like memory.

Chip manufacturers have sustained Moore's Law for a long time now: current

circuit components are about 65 nm wide.

Experts say this needs to shrink to 45 nm by 2009 and 22 nm by 2015. "Current semi-conductor technology is rapidly approaching its practical limits," says John E Kelly III, IBM's senior vice president of technology and intellectual property. "New, nanotechnology-based technologies will be needed to sustain the growth

that the IT industry provides to the world economy."

Indeed, it is today generally believed that to keep Moore's Law going—and possibly even redefine it—the field of nanotechnology will have to deliver. Recall *Move Over Silicon*, *Digit*, May 2006: transistors may soon use carbon nanotubes instead of silicon. The dedication of a research centre with a 70-teraflop supercomputer specially for such research seems a defining moment. ■